Thermal Analysis for IC Package Support

Engineering Reference — Package-Level Thermal Qualification

Objective

Characterize and validate the thermal performance of an integrated circuit (IC) package to ensure the junction temperature remains within its rated operating temperature across worst-case power and environmental conditions, and to support reliable, cost-effective package and board-level design decisions.

Scope

Covers steady-state and transient thermal simulation of the die, package (mold compound, substrate, lead frame/BGA balls, die-attach and TIM layers), and immediate board/enclosure environment, from early package selection through design verification.

Methodology

- Build a detailed CAD/mesh model of the die, package stack-up, PCB (copper layers, vias), and enclosure/heatsink where applicable.
- Apply JEDEC standard test environments (JESD51 series: natural/forced convection, 1s/2s board configurations) or application-specific boundary conditions.
- Run steady-state conduction/convection/radiation analysis (FEA/CFD) to extract Tj, θJA, θJB and θJC.
- Run transient analysis for power-cycling, burst, or pulsed-load scenarios to capture peak junction temperature and thermal time constants.
- Perform sensitivity/parametric studies on die size, copper spreading, TIM conductivity, and airflow to identify design margin and optimization paths.
- Correlate simulation results against bench measurements (thermocouple, IR imaging, or electrical Tj sensing) to validate the model.

Key Parameters & Targets

Parameter	Symbol / Metric	Typical Target
Junction Temperature	Tj (°C)	≤ 125°C (for silicon device spec)
Junction-to-Ambient Resistance	θJA (°C/W)	Per JEDEC JESD51 test board
Junction-to-Case Resistance	θJC (°C/W)	Per JESD51-14
Package Power Dissipation	Pd (W)	From device power budget
Board/Ambient Temperature	Ta (°C)	Application-defined (e.g., -55 – 85°C)

Thermal Model Creation Capabilities

- Detailed (DTM) and compact (CTM) package thermal models per DELPHI / JEDEC JESD15-4 methodology for fast board-level simulation.
- Support for QFN, BGA, LGA, CSP, flip-chip, SiP, and stacked-die/2.5D-3D package architectures.
- Automated CAD defeaturing and mesh generation from ECAD/MCAD source data (die, substrate, bond wires, balls/leads, mold compound).
- Two-resistor and DELPHI compact model extraction with boundary-condition independence for reuse across customer board designs.
- Model export/integration with industry tools: Siemens Simcenter Flotherm/FLOEFD, Ansys Icepak, 6SigmaET, and SPICE-based thermal networks.
- Model correlation and accuracy verification against physical test data (thermal test die, IR/thermocouple measurement).

Deliverables

- Thermal simulation report with θJA/θJC, Tj vs. power curves, and temperature contour plots.
- Recommended package, PCB, and system-level thermal mitigation (copper area, thermal vias, heatsink/TIM selection).
- Correlation summary between simulated and measured results, with model confidence assessment.

Standards Referenced

JEDEC JESD51 (thermal test methods), JESD51-2/-2A (natural convection), JESD51-6 (forced convection), JESD51-14 (θJC via cold plate), and JEP143 (junction temperature measurement).

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